

Dimensionless SG parameter result

Dimensionless geometric Savitzky-Golay experiment

Fixed reference scales

mass_kg: 2.3515975908123767
length_m: 0.19158499426207065
time_s: 0.13977207332379268
force_n: 23.061294513940194
torque_nm: 4.418197977129154
dimensionless_gravity: 1.0

Selected deterministic point (prior-free)

objective cost: 0.00027485441194423626
rotor delay [s]: 0.080859124928
gimbal delay [s]: 0.034159434568
optimizer success: True
optimizer message: three consecutive accepted steps satisfied relative objective-reduction tolerance
exact scale gauge: KKT hard constraint
unknown weak-mode cutoff: none
Jacobian rank: 13
Jacobian nullity: 1
common-scale analytic null norm: 5.403549776765736e-15
common-scale FD null norm: 1.0846758402832342e-09

Dimensionless coordinate

log_mass_scale	-1.31862907125
log_second_moment_mode_xx	0.553653423413
log_second_moment_mode_yy	2.69514877164
log_second_moment_mode_zz	8.38556126792
log_second_moment_mode_xy	-0.889758048473
log_second_moment_mode_xz	0.880907767301
log_second_moment_mode_yz	7.73445702063
dimensionless_cog_displacement_x	-0.386468371302
dimensionless_cog_displacement_y	4.92958653278
dimensionless_cog_displacement_z	0.0054670997062
log_force_effectiveness_1	-0.778625575383
log_force_effectiveness_2	-1.36517647536
log_force_effectiveness_3	-6.47469364343
log_force_effectiveness_4	-1.69723869754

Physical representative

mass [kg]: 0.6290565333408966
CoG body [m]: [-0.07606625 0.94440428 0.01055716]
force effectiveness: [0.45903649 0.25533561 0.00154197 0.18318867]
inertia [kg m²]:
[2146.66883, -74.43276532, -10.62086865]
[-74.43276532, 45.53081115, -299.9027399]
[-10.62086865, -299.9027399, 2106.697352]
principal moments [kg m²]: [1.47592109e-01 2.14930162e+03 2.14944778e+03]

Per-bag raw residual-wrench RMS

failure_2_single: [6.93930491e-02 8.67045242e-02 6.93557237e-02 2.56431391e+02
5.35914577e+01 3.53040136e+02]